Design and Analysis of Protograph LDPC-Coded PD-NOMA Systems With Symmetric-Hamming Mapping

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***Abstract*— This letter investigates the joint design of constellation mapping and protograph low-density parity-check (PLDPC) code for power-domain non-orthogonal multiple access (PD-NOMA) downlink systems. Specifically, we first propose a multi-priority design method to generate a type of symmetric-Hamming (SH) constellation mappings, and then verify their merits through a modified harmonic mean analysis. Furthermore, we construct a new type of non-punctured and punctured PLDPC codes by a mutual-information-guided threshold estimation to resist the inter-user interference. Simulation results demonstrate that the proposed designs achieve better performance than the existing benchmark schemes over Rayleigh fading channels.**

***Index Terms*— Non-orthogonal multiple access, constellation mapping, harmonic mean, protograph low-density parity-check codes, mutual information.**

I. Introduction

COMPARED to orthogonal multiple access (OMA), which permits the transmission of only one signal over a given resource, the power-domain non-orthogonal multiple access (PD-NOMA) enhances spectral efficiency by allowing multiple users to simultaneously share the same time and frequency resources, and thus is a promising multi-user access technology in future wireless networks [1], [2]. However, the superposition of multiple signals on a single resource inevitably introduces inter-user interference. To address this drawback, constellation mapping for superimposed signals has been investigated [3]. Also, the PD-NOMA performance has been further improved by establishing a correlation among mappings of multiple users [4]. In parallel, successive interference cancellation (SIC) has been employed at receiver to suppress inter-user interference [5]. Nevertheless, the effectiveness of SIC depends significantly on power allocation, as insufficient power disparities among users will impair a reliable symbol detection, resulting in residual interference and error propagation, which ultimately degrade the overall system performance [6].

Error-correcting codes (ECCs) are another key technology to resist the inter-user interference [7], [8]. Among various ECCs, low-density parity-check (LDPC) codes have been widely used for NOMA, especially the irregular LDPC codes, which can enhance NOMA performance at the cost of relatively high encoding complexity [9]. A class of structured LDPC codes, known as protograph LDPC (PLDPC) codes [10], [11], are a more desirable alternative for NOMA systems, as they maintain the performance advantages while enabling simpler encoding and decoding processes [8]. However, the convergence behavior of PLDPC codes in the NOMA scenario differs significantly from that in the single-user setting, motivating the development of dedicated multi-user coding schemes [7], [8].

Although various studies have explored the joint design of constellation mapping and ECCs for NOMA, there remains room for performance enhancement. Specifically, (i) For the existing NOMA, the modulation signals of different users are generated independently and then are simply superimposed based on power allocation [6], without fully exploiting the structural correlations among multiple user signals. (ii) Most existing multi-user codes designed for non-feedback scenarios cannot provide desirable extrinsic information in the iterative decoding (ID), and thus their iterative gain is limited. Motivated by the above observations, we conduct an investigation on the design of PLDPC-coded PD-NOMA systems. The main contributions of this letter are summarized as follows.

- We consider a PLDPC-coded PD-NOMA framework and propose a type of symmetric-Hamming (SH) mappings. The proposed SH mappings have a great flexibility in the bit-to-symbol mapping procedure and can effectively exploit soft information, thereby enhancing the overall system performance.
- Based on a modified protograph extrinsic information transfer (PEXIT) analysis, we design a new type of enhanced non-punctured and punctured PLDPC codes with the lowest decoding thresholds, thereby achieving the strongest robustness against the inter-user interference in the SH-mapped PD-NOMA system.
- Numerical results demonstrate that the proposed designs significantly outperform the existing counterparts over Rayleigh fading channels in terms of Harmonic mean, decoding threshold, and bit error rate (BER).

II. Conventional PD-NOMA System Model

Fig. 1 depicts the PD-NOMA downlink system consisting of a base station (BS) and two user equipments (UEs). At the BS (i.e., transmitting side), the information-bit sequence of UE-$k(k = 1, 2)$ $\mathbf{b}^k = \{b_1^k, b_2^k, \ldots, b_u^k\}$ is first passed to its corresponding UE-k PLDPC encoder, which then outputs a coded-bit sequence $\mathbf{g}^k = \{g_1^k, g_2^k, \ldots, g_w^k\}$. Subsequently, these two coded-bit sequences are interleaved individually, and then every m consecutive coded bits are converted into an M-ary modulation symbol. Specifically, the UE-1 and UE-2

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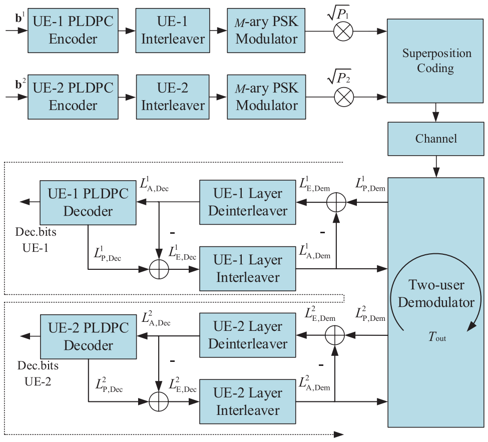


Fig. 1. Block diagram of the conventional PD-NOMA system.

modulators employ the high-power and low-power M-ary phase shift keying (PSK) mappings, respectively. In other words, both UE-1 and UE-2 modulation symbols (i.e., s^1 and s^2) are subject only to an energy constraint, where $s^k \in \mathcal{S} = \{s_i^k,\ i = 1, 2, \ldots, M\}$. We assume that the UE-$k$ transmission power is given as $P_k (k = 1, 2)$, where $P_1 = (1 - \alpha)P$, $P_2 = \alpha P$, α is the power allocation factor (PAF), and P is the total transmission power. Consequently, the modulation symbols from two different users are directly superimposed to produce the resultant transmission symbol as [1]

$$x_{\text{sc}} = \sqrt{P_1} s^1 + \sqrt{P_2} s^2. \tag{1}$$

After that, the received signal can be expressed as $y = hx_{\text{sc}} + n$, where h is the channel fading coefficient and $n \sim \mathcal{CN}(0, N_0)$ is the additive white Gaussian noise (AWGN). Subsequently, a turbo decoding framework composed of demodulator and decoder is used to recover the original information.

Given the received signal y and the a-priori demodulation log-likelihood ratios (LLRs) $\mathbf{L}_{\text{A,Dem}} = [\mathbf{L}^1_{\text{A,Dem}}, \mathbf{L}^2_{\text{A,Dem}}]$, the max-sum approximation of log-domain maximum a-posteriori probability (Max-Log-Map) algorithm is performed to calculate the extrinsic demodulation LLRs $\mathbf{L}^k_{\text{E,Dem}}$. In particular, the a-posteriori LLR for the l-th coded bit of UE-k during the two-user demodulation process is given by

$$L^{k,l}_{\text{P,Dem}} = \log \frac{\sum_{x_{\text{sc}} \in \chi_l^{(0)}} p(y|x_{\text{sc}}) \Pr(x_{\text{sc}}|\mathbf{L}_{\text{A,Dem}})}{\sum_{x_{\text{sc}} \in \chi_l^{(1)}} p(y|x_{\text{sc}}) \Pr(x_{\text{sc}}|\mathbf{L}_{\text{A,Dem}})}, \tag{2}$$

where $p(y|x_{\text{sc}})$ is the conditional probability density (CPD) function of received signal y given the transmission symbol x_{sc}, $\mathbf{L}^k_{\text{A,Dem}} = [L^k_{a,1}, L^k_{a,2}, \ldots, L^k_{a,m}]$ is the a-priori LLRs of coded bits from UE-k, $m = \log_2 M$, and χ_l^b denotes the joint constellation subset with the l-th coded bit being $b \in \{0, 1\}$. Also, $\Pr(x_{\text{sc}}|\mathbf{L}_{\text{A,Dem}})$ is given as

$$\Pr(x_{\text{sc}}|\mathbf{L}_{\text{A,Dem}}) = \prod_{l=1}^{2m} \Pr(b_l \mid L_{a,l}), \tag{3}$$

where the CPD function $\Pr(b_l \mid L_{a,l})$ is calculated as

$$\Pr(b_l \mid L_{a,l}) = \frac{\exp[(1 - b_l)L_{a,l}]}{1 + \exp(L_{a,l})}. \tag{4}$$

The extrinsic demodulation LLR $\mathbf{L}^k_{\text{E,Dem}} (k = 1, 2)$ will be used as the a-priori decoding LLR $\mathbf{L}^k_{\text{A,Dec}}$ for the UE-k decoder, which exploits the belief propagation (BP) algorithm to compute the corresponding extrinsic decoding LLR $\mathbf{L}^k_{\text{E,Dec}}$. This joint demodulation and decoding process repeats until both users are correctly decoded or the maximum number of outer iterations is reached. In this work, a bit-interleaved coded modulation with iterative decoding (BICM-ID) is considered.[1]

III. Proposed Symmetric-Hamming Mappings for PLDPC-Coded PD-NOMA System

A. Proposed SH Mapping

Constellation mapping plays an important role in optimizing PD-NOMA systems. Especially in the case of imperfect SIC, an excellent mapping mitigates inter-user interference, improves error-rate performance, and contributes to energy efficiency. Also, it facilitates flexible scheduling of quality of service (QoS) across different layers, ensuring the satisfaction of differentiated requirements and enhancing the overall system robustness [4]. Therefore, we establish a desirable correlation between two different users, and then generate a novel joint constellation mapping, called *SH mapping*. In the following, the design details for arbitrary higher-order SH mappings is illustrated.

The proposed SH mapping is composed of two M-PSK constellations corresponding to two different users. For clarity, given an M-PSK constellation for UE-k $(k = 1, 2)$, we define its M different PSK symbols as s_i^k $(i = 1, 2, \ldots, M)$ in a counterclockwise direction, with the initial point s_1^k located at the positive real axis. Furthermore, we define $\mathcal{N}_n = \sqrt{P_1} s_i^1 + \sqrt{P_2} s^2_{\pi_i(j)}$ as the i-th sub-constellation within the joint constellation, where $i \in \{1, 2, \ldots, M\}$ and $j \in \{1, 2, \ldots, M\}$. Then, the inter-constellation interference produced by two M-PSK constellations corresponding to two different users is the dominant factor degrading the error performance of the resultant joint constellation [4]. Therefore, we aim to mitigate this inter-constellation interference by minimizing the Hamming distance between two different sub-constellation labels with the smallest Euclidean distance, in other words, by maximizing the harmonic mean of the resultant joint constellation (i.e., achieve a desirable performance in the ID scenarios).

To be specific, in the proposed SH mapping, the permutation function $\pi_i(j)$ denotes the index of the j-th PSK symbol of UE-2 when UE-1 transmits its i-th PSK symbol. In other words, $\pi_i(j)$ determines the two symbols from two different users' constellations, and thus directly determines the joint symbol of the resultant joint constellation. Based on different permutation functions (i.e., achieving different UE-2 constellations for a given UE-1 constellation), the Hamming distances between two labels within any one sub-constellation will be different, which in turn influences the inter-constellation interference. Therefore, the proposed SH mapping must hold an optimized permutation function, and then can enable an appropriate joint constellation that establishes a favorable inter-user correlation and mitigate the inherent inter-constellation interference.

[1]The BICM-ID computes LLRs by enumerating M^2 constellation points, leading to a high demodulation complexity that scales as $\mathcal{O}(M^2 \log_2(M^2))$. However, the BICM-ID enables a reliable demodulation, which enhances the reliability of soft information provided to decoder and helps reduce the decoding complexity. Then, the BICM-ID implementation is practically feasible.

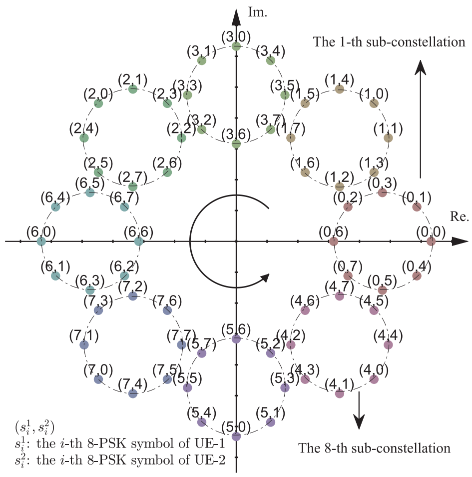


Fig. 2. The proposed SH mapping for two users with 8-PSK constellations.

To further clarify the design of the proposed SH mapping, three critical mathematical operators are defined as follows. We first suppose $\pi_i(j) = [a_{i,1}, a_{i,2}, \ldots, a_{i,M}]$, where $a_{i,j}$ is the label of the j-th PSK symbol of UE-2 when UE-1 transmits its i-th PSK symbol. A reversal operator is defined as $\mathcal{R}(\cdot)$ used to achieve permutation inversion $\mathcal{R}(\pi_i(j)) = [a_{i,M}, a_{i,M-1}, \ldots, a_{i,1}]$, while a cyclic shift operator is defined as $\mathcal{C}^{\lambda}(\cdot)$ used to achieve permutation cyclic shift by moving the first λ labels to the end of the array, i.e., $\mathcal{C}^{\lambda}(\mathcal{R}(\pi_i(j))) = [a_{i,M-\lambda}, \ldots, a_{i,1}, a_{i,M}, a_{i,M-1}, \ldots, a_{i,M-\lambda+1}]$ with $0 \le \lambda < M$. Based on the above operations, all joint symbols of the joint constellation for two users can be determined.

The detailed generation procedure of the SH mapping is summarized in *Algorithm 1* and can be described as the following three steps: (i) We begin with the first sub-constellation and initialize its permutation as $\pi_1(j) = [1, 2, \ldots, M]$. (ii) For the subsequent i-th ($i = 2, 3, \ldots, M$) sub-constellation, the cyclic shift parameter λ_i should be modified, while the corresponding permutation $\pi_i(j)$ is updated from $\pi_{i-1}(j)$ by applying $\mathcal{R}(\cdot)$ and $\mathcal{C}^{\lambda}(\cdot)$. (iii) After obtaining M distinct permutations $\{\pi_i(j)\}_{i=1}^{M}$, all M sub-constellations can be determined.

The resultant joint constellation (i.e., SH mapping) exploits the label correlation to reduce the Hamming distance between any two adjacent labels, thereby improving the overall system performance. As an example, Fig. 2 depicts the proposed SH mapping for two users with 8-PSK constellations. In particular, the constellation point $(1, 4)$ denotes the decimal representation of the label for both UE-1 and UE-2.

B. Harmonic Mean Analysis

Harmonic mean of the minimum squared Euclidean distance (HMMSED) is an effective criterion for analyzing constellation performance [12], [13]. However, the traditional harmonic mean analysis designed for single-user constellation is not applicable to joint constellation for two users. Therefore, we then adopt a modified harmonic mean analysis to validate the superiorities of the proposed SH mapping in non-feedback scenarios.

1) Modified HMMSED Analysis: For a joint constellation with M^2 labels, each label includes $2m$ bits, which is formulated by extracting m bits from UE-1 and m bits from UE-2. We define χ_l^b as the set of labels with the l-th bit being $b \in \{0, 1\}$, and $\chi_l^{\bar{b}}$ as the complement set of χ_l^b, where $l = 1, 2, \ldots, 2m$. Accordingly, the non-feedback HMMSED corresponding to the 1-st user (i.e., UE-1) is given as

$$D_h^1 = \left(\frac{1}{m 2^m} \sum_{l=1}^{m} \sum_{b=0}^{1} \sum_{x_{\text{sc}} \in \chi_l^b} \frac{1}{\|x_{\text{sc}} - z\|^2} \right)^{-1}, \quad (5)$$

where $z \in \chi_l^{\bar{b}}$ denotes the label closest to x_{sc}, and $\|x_{\text{sc}} - z\|$ denotes the Euclidean distance between x_{sc} and z. Similarly, the non-feedback HMMSED for UE-2 is given as

$$D_h^2 = \left(\frac{1}{m 2^m} \sum_{l=m+1}^{2m} \sum_{b=0}^{1} \sum_{x_{\text{sc}} \in \chi_{l,n}^b} \frac{1}{\|x_{\text{sc}} - z_{n'}\|^2} \right)^{-1}, \quad (6)$$

where $\chi_{l,n}^b$ is the subset of χ_l^b with labels belonging to the n-th sub-constellation, and $\|x_{\text{sc}} - z_{n'}\|$ denotes the Euclidean distance between x_{sc} and $z_{n'}$.

Remark 1: A larger non-feedback HMMSED can enhance the reliability of the initial soft information, thereby helping improve overall system performance [12]. As shown in Table I, the most significant position (MSP) mapping [14], optimized random mapping [12], and iterative-optimized mapping [12] yield identical D_h^1. Then, the disjoint-multilayer Gray (D-Gray) mapping [6] and proposed SH mapping achieve a larger D_h^1, indicating more superior UE-1 performance in the non-feedback scenario. Moreover, in the case of identical D_h^1, the proposed SH mapping has a larger D_h^2 than the D-Gray mapping, further demonstrating its overall performance advantage. Note that since the BICM-ID performance significantly relies on the BICM (i.e., non-feedback) performance, the non-feedback HMMSED serves as the dominant factor guiding optimization in both non-feedback and feedback scenarios [15].

Algorithm 1 Generation of the Proposed SH Mapping

Input: M-PSK symbols $s_i^k \in \mathcal{S}$ ($k = 1, 2$), power P_1, P_2.
Permutation Initialization and Update:
Initialize $\pi_1(j) = [1, 2, \ldots, M]$;
for $i = 2$ **to** M **do**
  Compute cyclic shift parameter:
  $$\lambda_i = (M - 2(i-1)) \bmod M.$$
  Update permutation function:
  $$\pi_i(j) = \mathcal{C}^{\lambda_i}\big(\mathcal{R}(\pi_{i-1}(j))\big).$$
Generate M-PSK + M-PSK constellation points:
for $i = 1$ **to** M **do**
  for $j = 1$ **to** M **do**
    $x_{i,j} = \sqrt{P_1} s_i^1 + \sqrt{P_2} s_{\pi_i(j)}^2$;
Output: SH constellation set $\chi = \{x_{i,j}\}_{i,j=1}^{M^2}$.

TABLE I

HMMSED Analyses for Different Constellation Mappings, Including the MSP Mapping, Optimized Random (Opti. Rand.) Mapping, Iterative-Optimized (Iter. Opt.) Mapping, D-Gray Mapping and the Proposed SH Mapping

HMMSED \ Mapping	MSP	Opt. Rand.	Iter. Opt.	D-Gray	SH
D_h^1	0.0623	0.0623	0.0623	0.0746	0.0746
D_h^2	0.0620	0.0799	0.0837	0.0725	0.2477

IV. Proposed PLDPC Codes for PD-NOMA System With SH Mapping

A protograph with code rate $R = (q_v - q_c)/q_v$ is represented by a base matrix $\mathbf{B} = (\delta_{p,r})$ of size $q_c \times q_v$, where q_c and q_v denote the number of check nodes (CNs) and variable nodes (VNs), respectively, and $\delta_{p,r}$ denotes the number of edges between the p-th CN and the r-th VN. With the aid of a modified version of PEXIT algorithm [16], we design two new PLDPC codes tailored for the proposed SH-mapped PD-NOMA system. It is worth noting that the code design takes into account the characteristic of proposed SH mapping so as to approach the achievable capacity. The proposed PLDPC-code deign procedure is illustrated in the following.

A. Unpunctured Code

For the single-user PLDPC code, the proportion of degree-2 VNs is typically kept below 50% to achieve a good waterfall performance while avoiding the error floor at high signal-to-noise ratio (SNR) [10]. However, such a design guideline is not applicable in multi-user ID scenarios. To be specific, each transmitted symbol carries coded bits from two different users, and then the demodulation for any one user can benefit from the soft information provided by another user during outer iterations. Obviously, such a cross-user information exchange mitigates the negative impact on the error-floor behavior typically associated with a high proportion of degree-2 VNs. Motivated by this property, we increase the proportion of degree-2 VNs to around 50%, exceeding the conventional design in single-user communications.

A rate-$1/2$ TU-1 code corresponding to a 3×6 base matrix is confined as: (i) set a highest-degree VN with a degree no greater than 6, i.e., $\delta_{1,1} + \delta_{2,1} + \delta_{3,1} \leq 6$; (ii) initialize the number of degree-2 VNs $p_{v,2} \geq 3$, i.e., $\delta_{1,k} + \delta_{2,k} + \delta_{3,k} = 2$ for $k = 2, 3, 4$; (iii) reduce the code-search space by constraining the remaining VNs to have degrees no greater than 3, i.e., $\delta_{1,k} + \delta_{2,k} + \delta_{3,k} \leq 3$ for $k = 5, 6$; (iv) lower the encoding and decoding complexity by constraining all elements to be no greater than 3, i.e., $\delta_{p,r} \in \{0, 1, 2, 3\}$ for $p = 1, 2, 3$ and $r = 1, 2, \ldots, 6$. Consequently, both initial and resultant base matrices (i.e., $\mathbf{B}_{\mathbf{U}}^{\mathbf{0}}$ and $\mathbf{B}_{\mathbf{U}}$) are illustrated as

$$\mathbf{B}_{\mathbf{U}}^{\mathbf{0}} = \begin{bmatrix} \delta_{1,1} & 0 & \delta_{1,3} & \delta_{1,4} & \delta_{1,5} & \delta_{1,6} \\ \delta_{2,1} & 1 & \delta_{2,3} & \delta_{2,4} & \delta_{2,5} & \delta_{2,6} \\ \delta_{3,1} & 1 & \delta_{3,3} & \delta_{3,4} & \delta_{3,5} & \delta_{3,6} \end{bmatrix},$$
$$\mathbf{B}_{\mathbf{U}} = \begin{bmatrix} 2 & 0 & 1 & 1 & 0 & 2 \\ 2 & 1 & 0 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 & 2 & 1 \end{bmatrix}. \tag{7}$$

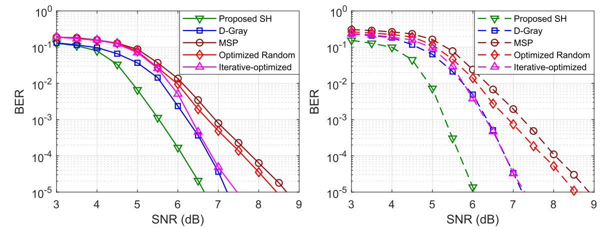
B. Punctured Code

Additionally, we further present a rate-$1/2$ punctured PLDPC code, namely TU-2 code, whose 3×5 base matrix holds the following constraints: (i) has a pre-coding structure [10] composed of a degree-1 VN-1, a highest-degree punctured VN-2, and a associated CN only connecting VN-1 and VN-2; (ii) the degree of VN-2 is not greater than 5, i.e., $\delta_{1,2} + \delta_{2,2} + \delta_{3,2} \leq 5$, while the degree of the remaining VN-3$\sim$-5 are not greater than 3, i.e., $\delta_{2,k} + \delta_{3,k} \leq 3$ for $k = 3, 4, 5$; (iii) all elements in the base matrix are not greater than 3, i.e., $\delta_{p,r} \in \{0, 1, 2, 3\}$ for $p = 1, 2, 3$ and $r = 1, 2, \ldots, 5$.

TABLE II

Decoding Thresholds of Both Users With the (3, 6), RJA, T-PLDPC, E-PLDPC, Proposed TU-1 and TU-2 Codes Under SH Mapping Over a Rayleigh Fading Channel. The Maximal Outer and Inner Iterations Are 4 and 50, Respectively

Thresholds	TU-1 Code	TU-2 Code	regular-(3, 6) [10]
γ_{th}^{1}(dB)	4.282	4.370	4.891
γ_{th}^{2}(dB)	5.750	5.488	6.094
Thresholds	RJA [11]	T-PLDPC [7]	E-PLDPC [8]
γ_{th}^{1}(dB)	4.885	4.620	4.561
γ_{th}^{2}(dB)	5.896	5.870	5.757



(a) (b)

Fig. 3. BER performance of the regular-(3, 6) PLDPC-coded PD-NOMA system with QPSK modulation over a Rayleigh fading channel. In particular, the D-Gray, MSP, optimized random, iterative-optimized, and proposed SH mappings are considered: (a) UE-1; (b) UE-2.

Similarly, both initial and resultant base matrices (i.e., $\mathbf{B}_{\mathbf{P}}^{\mathbf{0}}$ and $\mathbf{B}_{\mathbf{P}}$) are illustrated as

$$\mathbf{B}_{\mathbf{P}}^{\mathbf{0}} = \begin{bmatrix} 1 & 2 & 0 & 0 & 0 \\ 0 & \delta_{2,2} & \delta_{2,3} & \delta_{2,4} & \delta_{2,5} \\ 0 & \delta_{3,2} & \delta_{3,3} & \delta_{3,4} & \delta_{3,5} \end{bmatrix},$$
$$\mathbf{B}_{\mathbf{P}} = \begin{bmatrix} 1 & 2 & 0 & 0 & 0 \\ 0 & 1 & 1 & 2 & 1 \\ 0 & 1 & 2 & 1 & 1 \end{bmatrix}, \tag{8}$$

where the second column represents the punctured VN-2.

Remark 2: The PLDPC-code design takes into account the employed bit-to-symbol mapping (i.e., SH mapping), especially in the ID scenarios where the soft information exchanged between demodulator and decoder plays an important role in the coded-modulation system. Accordingly, the proposed PLDPC codes are tailored to the PD-NOMA system by jointly considering the SH mapping and the ID mechanism.

V. Numerical Results

A. PEXIT Analysis

A lower decoding threshold implies a faster convergence of the BP decoding, thereby enabling a reliable communication at a lower SNR region. As shown in Table II, we compare the decoding thresholds between our codes and existing codes (i.e., regular-$(3, 6)$ [10], repeat-jagged accumulate (RJA) [11], T-PLDPC [7] and E-PLDPC [8]) in the SH-mapped PD-NOMA system with 8-PSK modulation over a Rayleigh fading channel. Unless otherwise specified, all codes have a codeword length of 1008 and a code rate of $1/2$, while γ_{th}^{k} denotes the decoding threshold of UE-k ($k = 1, 2$). The PEXIT analyses indicate that the proposed TU-2 code achieves threshold gains of approximately 0.19 dB and 0.26 dB for the first and second users, respectively, compared with the E-PLDPC code, which outperforms the regular-$(3, 6)$, RJA, and T-PLDPC codes. Moreover, the proposed TU-1 code provides an additional gain of about 0.1 dB in the decoding threshold of the first user compared to the proposed TU-2 code.

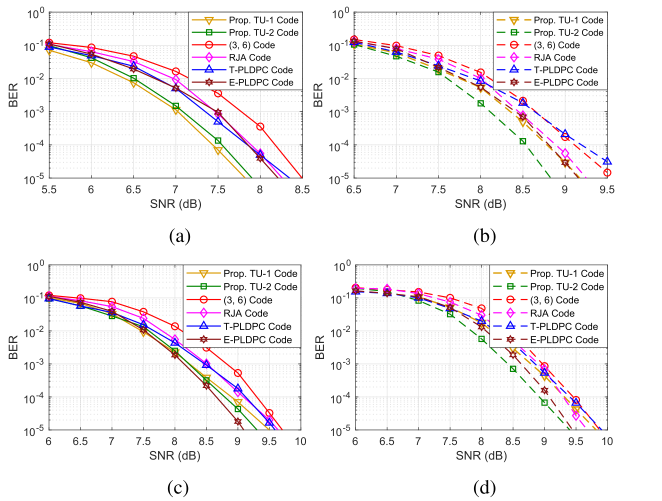


Fig. 4. BER performance of the SH-mapped PD-NOMA system with 8-PSK modulation over a Rayleigh fading channel. The regular-(3, 6), RJA, T-PLDPC, E-PLDPC, and proposed TU-1 and TU-2 PLDPC codes are adopted: (a) UE-1, $T_{\text{out}} = 4$; (b) UE-2, $T_{\text{out}} = 4$; (c) UE-1, $T_{\text{out}} = 0$; (d) UE-2, $T_{\text{out}} = 0$.

B. BER Performance

In this section, we present the BER simulations to validate the superiorities of proposed SH mapping and two new PLDPC codes in PD-NOMA system over a Rayleigh fading channel. In particular, both QPSK and 8-PSK modulation schemes are adopted, with the corresponding values of α set to 0.3 and 0.2, respectively, according to the minimum average BER criterion. Also, the maximum numbers of outer and inner iterations are set to $T_{\text{out}} = 4$ and $T_{\text{in}} = 50$, respectively. In particular, $T_{\text{out}} = 4$ is selected since the BER performance nearly converges at this point, offering a balanced tradeoff between error performance and computational complexity.

Fig. 3 compares the BER performance of the D-Gray, MSP, optimized random, iterative-optimized, and our proposed SH mappings. As observed from this figure, the proposed SH mapping achieves a superior BER performance compared with the other four existing mappings. At a BER of 1×10^{-5}, the proposed SH mapping provides performance gains of about $0.5 \sim 2.0$ dB for UE-1 and $1.1 \sim 2.8$ dB for UE-2 compared to these benchmark mappings.

Fig. 4 compares the BER performance of the regular-$(3, 6)$ [10], RJA [11], T-PLDPC [7], E-PLDPC [8], and proposed TU-1 and TU-2 codes in the PD-NOMA system with 8-PSK modulation. Obviously, the proposed TU-1 and TU-2 codes outperform the other four PLDPC codes in an ID scenario. Specifically, at a BER of 1×10^{-5}, the proposed TU-1 code and TU-2 code with $T_{\text{out}} = 4$ achieve iterative gains of around 1.7 dB and 1.5 dB, respectively, at UE-1 compared with the case of $T_{\text{out}} = 0$, and iterative gains of around 0.5 dB for both codes at UE-2 under the same condition. Additionally, compared with the best-performing E-PLDPC code in the ID scenario, the proposed TU-2 code achieves gains of approximately 0.3 dB for both UE-1 and UE-2, whereas the proposed TU-1 code attains a gain of about 0.4 dB for UE-1 without degrading the performance of UE-2.

Another important measure for PLDPC codes is decoding complexity expressed as $\mathcal{O}\left(T_{\text{in}}\left(u\bar{V} + (w-u)\bar{C}\right)\right)$ [17], where $\bar{V}$ (resp. $\bar{C}$) denotes the average degree of VNs (resp. CNs), w and u denote the codeword length and information length, respectively. Based on the base matrices of TU-2 and E-PLDPC codes, one can find that the proposed TU-2 code exhibits a lower decoding complexity $\mathcal{O}\left(T_{\text{in}}\left(5.198w\right)\right)$ than the E-PLDPC code with $\mathcal{O}\left(T_{\text{in}}\left(6.000w\right)\right)$.

VI. Conclusion

This letter has investigated the design of constellation mappings and PLDPC codes for PD-NOMA downlink system with two users. To be specific, we have proposed a multi-priority design method to develop a type of SH constellation mappings, and verified their merits from the perspective of harmonic mean analysis. Additionally, we have utilized an mutual-information-guided threshold estimation to construct both TU-1 and TU-2 codes with desirable convergence performance. Theoretical and simulated results have demonstrated that the proposed designs achieve better performance than state-of-the-art counterparts over Rayleigh fading channels.

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